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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

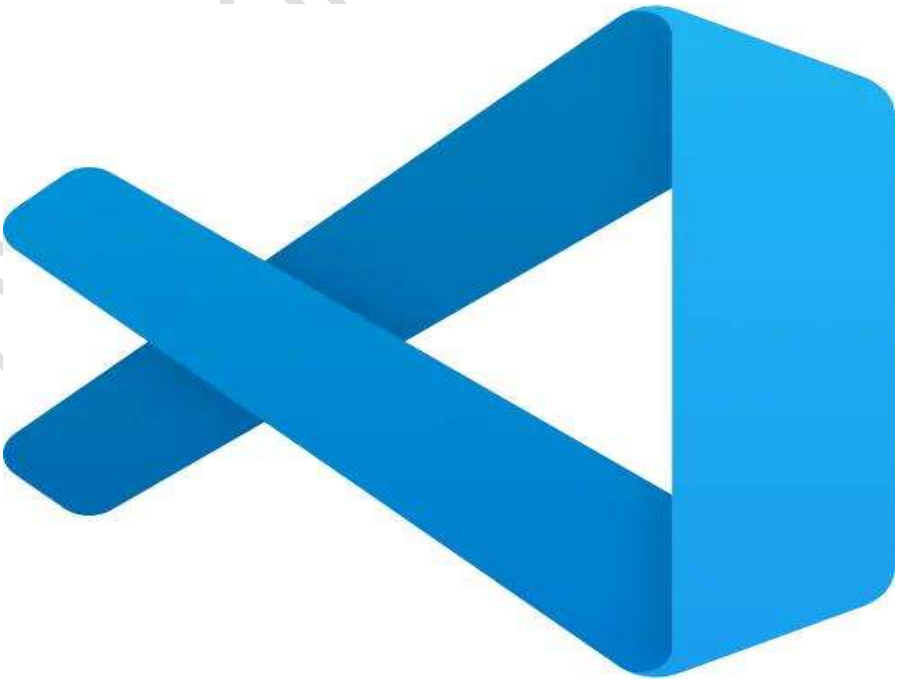
Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Lesson 13: VS Code Introduction



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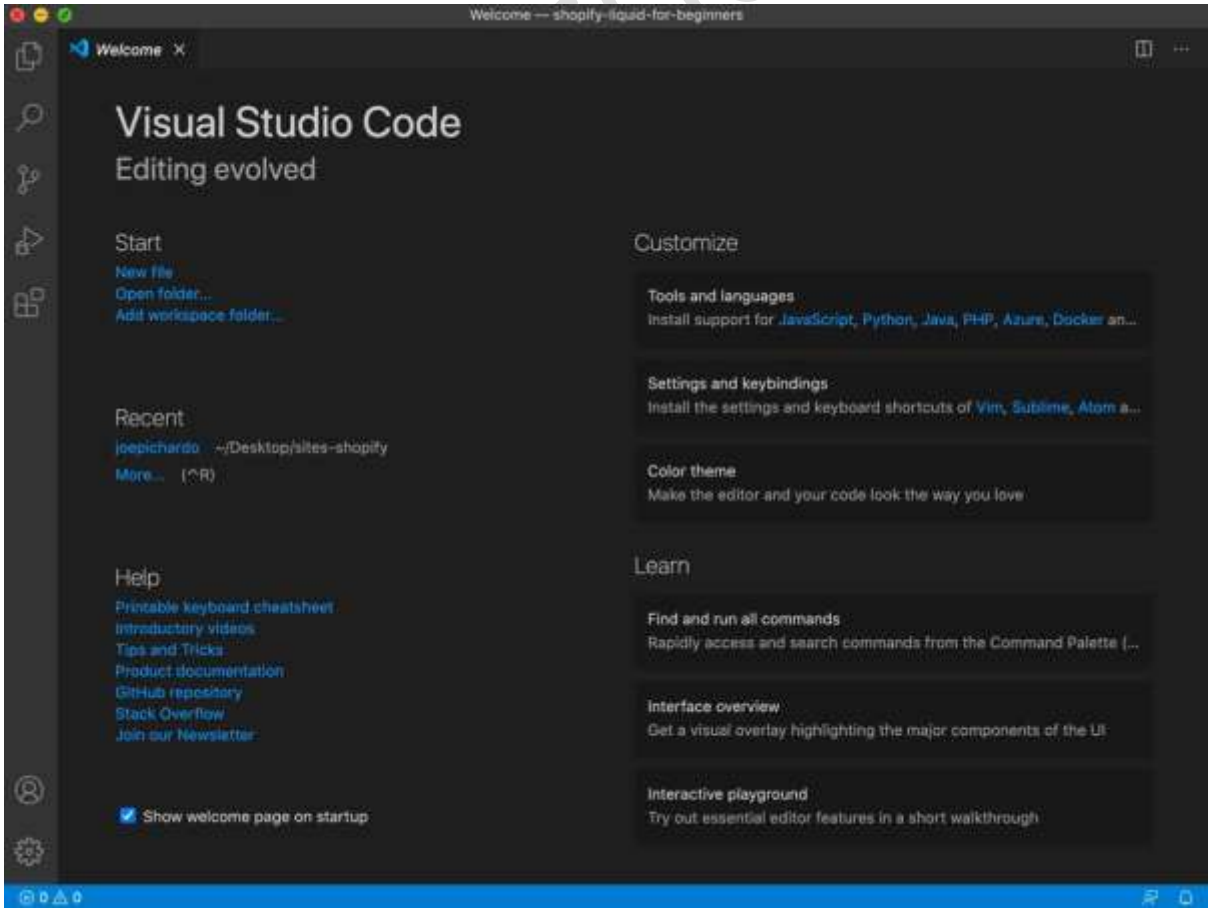
Learning Objectives

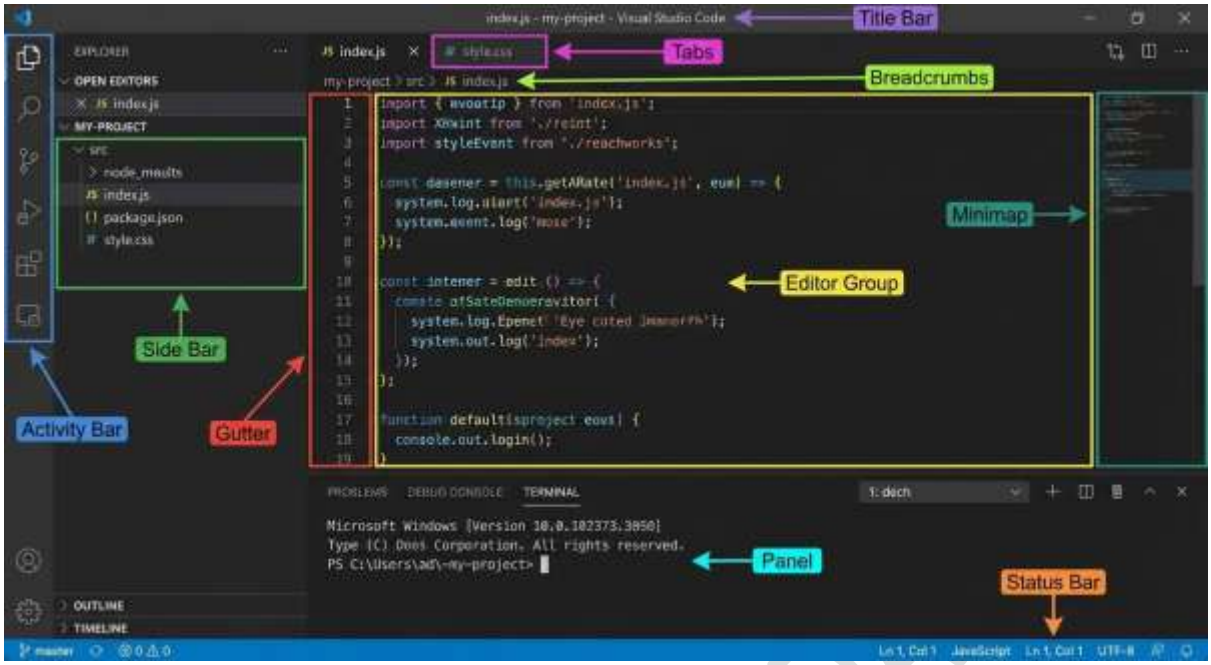
By the end of this lesson, students will be able to:

- Understand what VS Code is.
- Know why programmers use VS Code.
- Identify the main parts of the VS Code window.
- Create and save a Python file.
- Run a simple Python program in VS Code.
- Use basic keyboard shortcuts.

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What is VS Code?





VS Code stands for **Visual Studio Code**.

It is a **free code editor** developed by **Microsoft**.

A code editor helps programmers write, edit, and organize computer programs.

Think of it like:

- 📖 Notebook → Writing notes
- 🎨 Paint → Drawing pictures
- 📝 VS Code → Writing computer programs

Why Do We Use VS Code?

VS Code helps programmers:

- ✍️ Write code easily
- 🎨 Highlight code with colors
- 📁 Organize project files
- 🐞 Find mistakes (errors)

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- ▶ Run programs
- 🔧 Install helpful extensions

Many professional programmers use VS Code every day.

Features of VS Code

✅ Free to use

✅ Easy to learn

✅ Fast

✅ Colorful coding

✅ Supports many programming languages

Examples:

- Python
- Java
- HTML
- CSS
- JavaScript
- C++
- PHP

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Installing VS Code

1. Open your web browser.
2. Visit the official VS Code website.
3. Click **Download for Windows**.

4. Open the downloaded installer.
5. Follow the setup instructions.
6. Click **Finish**.

Now VS Code is ready to use.

Opening VS Code

Click:

Start Menu

↓

Type:

Visual Studio Code

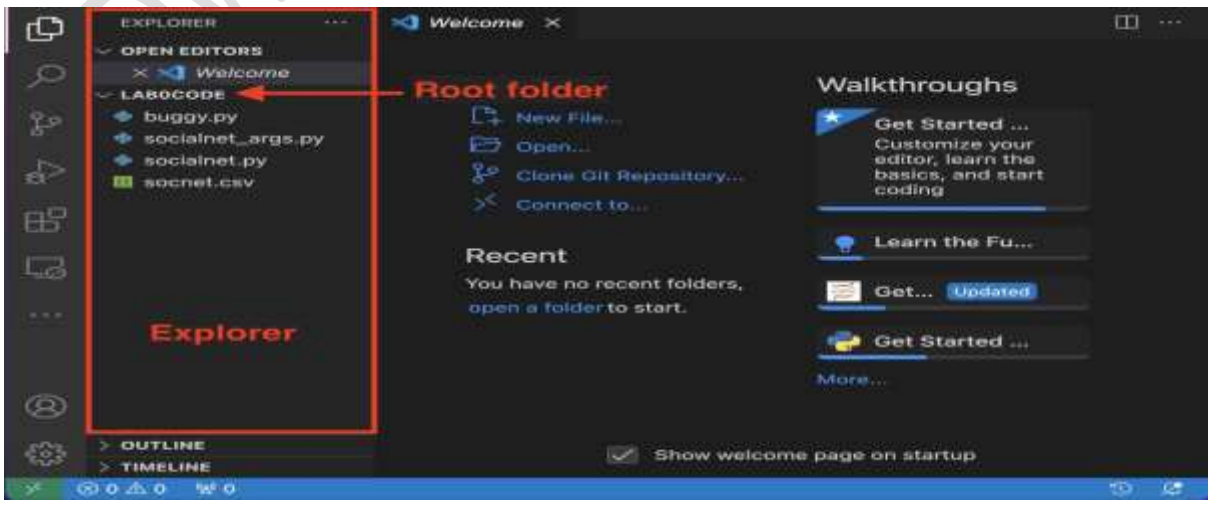
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Click:

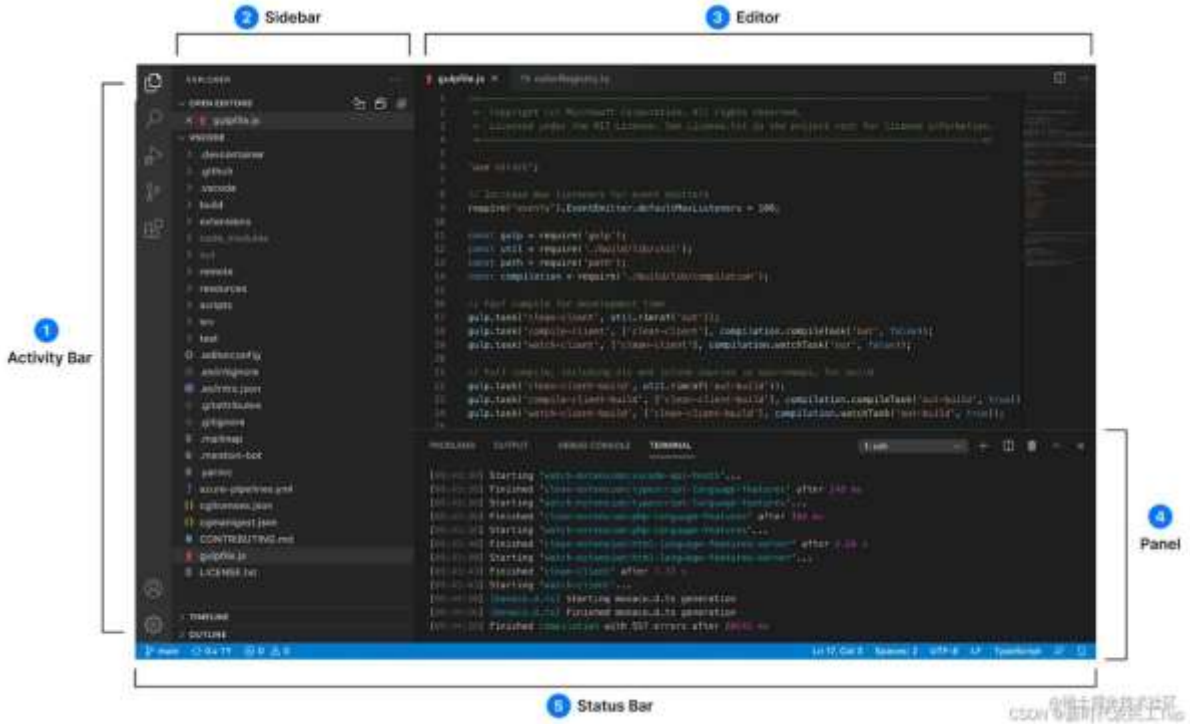
Visual Studio Code

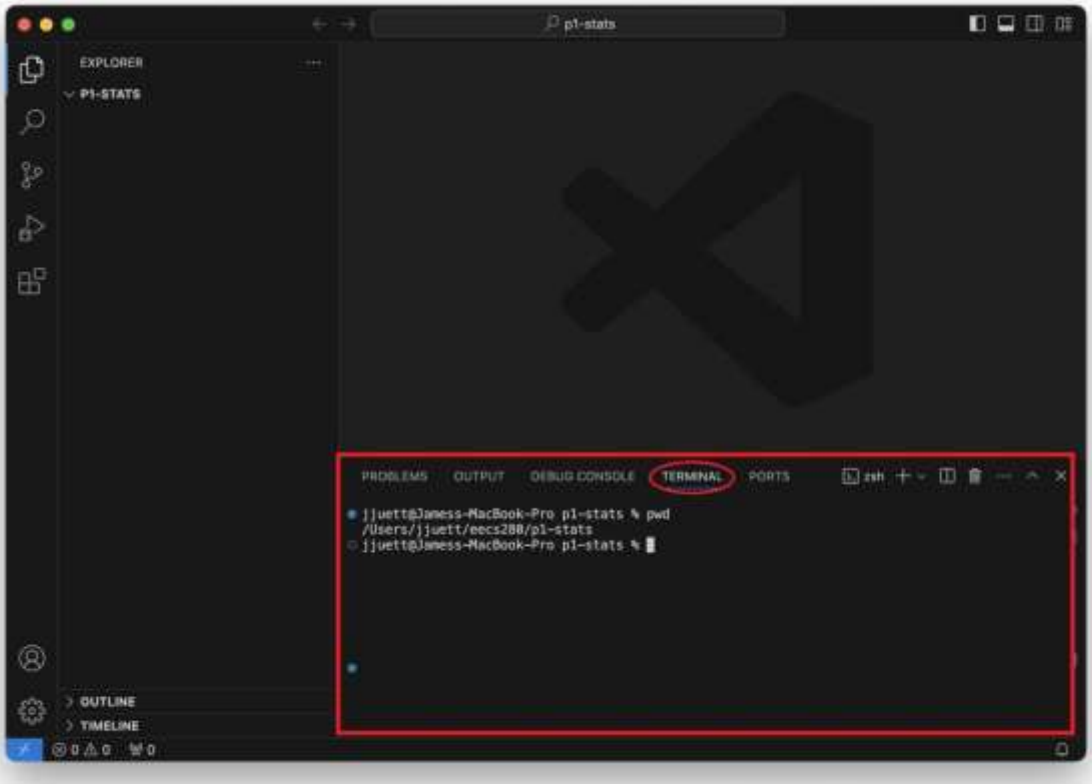
The VS Code window opens.

Parts of VS Code



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1 Activity Bar

Located on the left.

It contains icons for:

- Explorer
- Search
- Source Control
- Run
- Extensions

2️⃣ Explorer

Shows your project files and folders.

3️⃣ Editor

This is where you write your programs.

4️⃣ Terminal

The terminal runs your programs.

You can open it using:

Terminal → New Terminal

Shortcut:

Ctrl + `

5️⃣ Status Bar

Located at the bottom.

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It shows:

- Programming language
- Errors
- Notifications

Creating Your First Folder

Click

File → Open Folder

Create a folder named:

Python Practice

Click

Select Folder

Creating a Python File

Inside the folder

Click

New File

Type

hello.py

The extension **.py** tells the computer it is a Python program.

Writing Your First Program

Type:

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print("Hello Students!")

Save the file.

Shortcut:

Ctrl + S

Running the Program

Open Terminal.

Type:

python hello.py

Press Enter.

Output:

Hello Students!

Another Example

print("Welcome to VS Code")

Output

Welcome to VS Code

Student Activity

Write your own program:

print("My Name is _____")

print("I am learning Python")

Replace the blank with your name.

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Changing the Theme

VS Code allows different color themes.

Steps:

1. Click **File**
2. Preferences
3. Theme
4. Select a favorite theme

Examples:

- Dark Theme
- Light Theme

Extensions

Extensions add extra features to VS Code.

Popular Extension:

Python Extension

It helps by:

- Showing mistakes
- Running Python easily
- Giving code suggestions

Basic Keyboard Shortcuts

Shortcut Action

Ctrl + S Save File

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Shortcut	Action
Ctrl + N	New File
Ctrl + O	Open File
Ctrl + C	Copy
Ctrl + V	Paste
Ctrl + Z	Undo
Ctrl + Y	Redo
Ctrl + `	Open Terminal

VS Code vs Notepad

VS Code Features Compared with Notepad

Number of useful programming features available.

0369NotepadVS Code

Notepad	VS Code
Plain text	Colorful code
No extensions	Extensions available
No terminal	Built-in terminal
Basic editing	Smart coding tools
Simple text	Professional coding

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Practical Activities

Practical Activity

Students should:

1. Open VS Code.
2. Create a folder named **Python Practice**.
3. Create a file called **hello.py**.
4. Type:

```
print("Hello World")
```

5. Save the file.
6. Run the program.
7. Observe the output.

Classroom Activity

Ask students to:

- Open VS Code.
- Create their own Python file.
- Print their:
    - Name
    - Class
    - School
    - Favorite Subject

Example:

```
print("Name : Riya")
print("Class : 7")
```

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```
print("School : ABC High School")
print("Favorite Subject : Science")
```

Mini Challenge

Create a file called:

aboutme.py

Write:

```
print("My Name is Aarav")
print("I love Computers")
print("I want to become a Software Engineer")
```

Run the program successfully.

Important Terms

Word	Meaning
VS Code	Code editor
Editor	Place where code is written
Explorer	Shows files and folders
Terminal	Runs programs
Extension	Adds new features
Save	Store the file
Python File	File ending with .py

Common Mistakes

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❌ Forgetting to save the file before running.

❌ Naming the file as:

hello

instead of

hello.py

❌ Missing quotation marks:

Wrong:

print(Hello)

Correct:

print("Hello")

Worksheet

Fill in the Blanks

1. VS Code is a _____ editor.

Answer: Code

2. VS Code is developed by _____.

Answer: Microsoft

3. Python files end with _____.

Answer: .py

4. We use _____ to save a file.

Answer: Ctrl + S

5. The Terminal is used to _____ programs.

Answer: Run

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Quiz

1. What does VS Code stand for?

A. Virtual Studio Code

B. Visual Studio Code ✅

C. Video Studio Code

D. Visual Software Code

2. Who developed VS Code?

A. Google

B. Microsoft ✅

C. Apple

D. Meta

3. Which file extension is used for Python?

A. .txt

B. .doc

C. .py ✅

D. .jpg

4. Which shortcut saves a file?

A. Ctrl + P

B. Ctrl + S ✅

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C. Ctrl + R

D. Ctrl + T

5. Which part of VS Code is used to run programs?

A. Explorer

B. Activity Bar

C. Terminal ✅

D. Status Bar

Learning Outcomes

After completing this lesson, students will be able to:

- ✅ Explain what VS Code is.
- ✅ Identify the main parts of the VS Code window.
- ✅ Create and save a Python file.
- ✅ Run a simple Python program.
- ✅ Use basic keyboard shortcuts.
- ✅ Begin writing Python programs confidently in VS Code.

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